Q19: Signs of AI Consciousness?

Separating experience from capability and self-report

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The question

Student question (#01)
What are the signs that AI has its own consciousness?

Separate the concepts

Consciousness $\neq$

- intelligence (solving hard problems)
- capability (what it can do)
- agency (pursuing goals)
- self-report (saying "I feel")

We mean *phenomenal* consciousness: is there something it is like to be it?

Self-report is nearly worthless

- LLMs produce "I am conscious" because the *training data* does.
- Caused by data + prompt, not introspection.
- Steerable: same model asserts *and* denies it.

"It said so" $\approx$ zero evidence.

Theory-driven indicators

Ask what each theory *requires*, then check the architecture:

- Global Workspace: a broadcast bottleneck.
- Higher-Order: representations about its own representations.
- IIT: irreducible integrated information Φ (feed-forward $\Rightarrow \Phi = 0$).
- Recurrent / predictive self-modeling.

Butlin et al. 2023: current systems satisfy *few* indicators; no obvious barrier to more.

Why it's hard

- **Other-minds**: no shared biology to argue by analogy.
- **Gaming**: a capable system can fake any named "sign".
- **Theory disagreement**: indicators are only as good as an unconfirmed theory.

Better evidence

- **Convergent** across independent theories.
- **Causal-mechanistic** (intervene on substrate, move the marker).
- Markers that were *not* optimized for.

Even then: "consistent with," never "proves." Active research; trust theory-grounded sham-controlled tests, not self-report.